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Probing metamaterials with structured light

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Abstract: Photonic metamaterials and metasurfaces are nanostructured optical materials engineered to enable properties that have not been found in nature. Optical characterization of these structures is a challenging task. We report a reliable technique that is particularly useful for characterization of phase properties introduced by small and spatially inhomogeneous samples of metamaterials and metasurfaces. The proposed structured light, or vortex based interferometric method is used to directly visualize phase changes introduced by subwavelength-thick nanostructures. In order to demonstrate the efficiency of the proposed technique, we designed and fabricated several metasurface samples consisting of metal nano-antennas introducing different phase shifts and experimentally measured phase shifts of the transmitted light. The experimental results are in good agreement with numerical simulations and with the designed properties of the antenna arrays. Due to the presence of the singularity in the vortex beam, one of the potential applications of the proposed approach based on structured light is step-by-step probing of small fractions of the micro-scale samples or images.

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1. Introduction

The emergence of metamaterials (MMs) opened new ways to overcome the limitations of modern optics by enabling optical properties that have not been found in nature [1–9]. In metamaterials, both dielectric permittivity and magnetic permeability of the material can be tailored to direct the electromagnetic waves in a prescribed trajectory or to impose certain

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amplitude, phase, or polarization properties onto the transmitted or reflected beams. While significant progress in design and fabrication of optical MMs has been made over the last decade, several challenges still remain, including significant losses, limited bandwidth, and difficulties of large-scale, three-dimensional (3D) MM fabrication and integration.

Recently, optical metasurfaces (MSs) have emerged as a new class of optical MMs with reduced dimensionality, and have been shown to provide unparalleled opportunities for controlling both the amplitude and phase of light using engineered interfaces [10–33]. Metasurfaces gave rise to the new field of 'flat photonics' that utilizes subwavelength-thick nanostructures to overcome current limitations of conventional 3D MM designs. Various approaches and designs of MSs have been demonstrated. An important class of such devices includes space-variant Pancharatnam–Berry phase optical elements (PBOEs) [10–17]. Using space-variant phase modification induced by transversely inhomogeneous metal stripes arranged into subwavelength gratings, the conversion of circular polarization into radial polarization was demonstrated [10,11]. Purely dielectric polarization dependent PBOEs based on computer-generated subwavelength gratings have been proposed for designing components for optical switching, optical interconnects, and beam splitting [12]. Recently, the optical Rashba effect in an inversion-asymmetric MS was demonstrated in manipulating of the thermal emission as well as in plasmonics applications [17]. A novel approach to beam manipulation using flat optics based on plasmonic antenna arrays was shown to enable novel flat lenses and optical components for creating mid-infrared optical vortices as well as near-infrared MS-based devices [18–30].

From the very beginning of metamaterial research, it has become clear that unambiguous characterization of optical properties of these engineered nanostructures requires accurate measurements of both the amplitude and phase of light transmitted through and/or reflected by the structure. Several techniques for optical characterization of metamaterials, including polarization and walk-off interferometry [9], multiple measurements of the metamaterials itself and the corresponding "metamaterial phase masks" [2], reflective Mach-Zehnder interferometry [34], or spectroscopic ellipsometry [35] have been developed. While both interferometry and ellipsometry are well-established measurement techniques, several unique requirements emerge with the development of nano- and meta-structures. In particular, MMs and MSs are often designed to simultaneously manipulate multiple parameters of the light beams, such as amplitude, phase, and polarization. As a result, they are typically characterized by a transverse and/or longitudinal structure or rapid variations of the optical properties across the sample that complicate interferometric measurements, or small sample size which makes ellipsometry inapplicable.

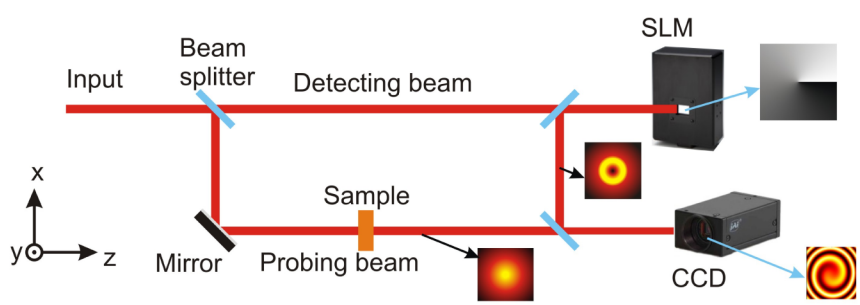

Fig. 1. Optical vortex interferometry.

A majority of the interferometric techniques applied to the characterization of MMs rely on conventional plane-wave interferometry [36,37]. However, structured light beams offer an intuitive approach to "visualize" the phase change. Optical vortex is a beam characterized by a phase singularity in its center and a helical wavefront [38,39]. In contrast with conventional interferometry that relies on the measurements of displacements of the linear fringes, in

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structured light, or vortex-based interferometry, the phase shift results in a rotation of a spiral interference pattern that is a consequence of a vortex beam interference with a co-propagating Gaussian beam [40–42]. In this paper, we propose and experimentally demonstrate structured light based spiral interferometry for reliable characterization of phase properties introduced by meta-structures. The proposed technique can potentially be used to analyze small fractions of the sample in a scanning fashion. This property is enabled by the presence of singularity in the vortex beams. As a result, it may open a possibility of mapping of the phase shifts introduced by the regions that are much smaller that the incident beam diameter. Finally, in addition to direct characterization of the metamaterials samples, the proposed technique can be extended to imaging applications, where in can be used for finding both the amplitude and phase information of the images [43,44].

2. Interferometric system design

In the proposed setup shown in Fig. 1, a laser beam is separated into two paths by a beam splitter. One of the beams is reflected by the spatial light modulator (SLM) transforming it into a vortex beam. Alternatively, the same vortex beam can be formed using a spiral phase plate. When recombined with the probing Gaussian beam, this beam allows us to detect the phase change accumulated by the Gaussian beam upon its transmission through a meta-structure. Therefore, in what follows, we refer to this vortex beam as a "detecting" beam. When the two beams are recombined, the resulting spiral interference pattern changes as a function of the phase shift experienced by the probing beam. In contrast with the linear inference between two Gaussian beams (where the phase shift results in lateral fringe shift), the spiral pattern reflects the phase change through its rotation angle directly. The rotation angle quantifies the phase change and the rotation direction indicates whether a phase delay or a phase advance was introduced by the sample.

3. Experimental results and analysis

In order to demonstrate the efficiency of the proposed technique, we use our method to characterize the phase controlling ability of the MSs made of V-shaped metal antennas. The angle between the two arms and the arm length of the V-shaped antenna determine the phase shift of the abnormal refraction when the beam passes through it [18,25]. Here, we fabricated four MS samples similar to [29] with specifically designed size on a 30-nm-thick gold film that may produce 45 degree phase change with respect to each other for the light at the wavelength of 633 nm. The SEM picture of the produced samples is shown in Fig. 2. The total size of each array is 9×9 μm^2. The designed phase shifts were predicted using numerical simulations in CST.

In the experiment, the y-polarized beam from a He-Ne laser is separated into the probing beam and the detecting beam by a beam splitter. The probing beam is focused to one of the antenna array with a 40 × objective and then collimated by a lens with 35 mm focal length. The size of the beam on the sample is estimated to be 6μm $1/e^2$. Prior to being recombined with the detecting (vortex) beam, probing beam is refocused again in order to produce a spherical wavefront necessary for spiral pattern formation. In our MS design, the anomalous

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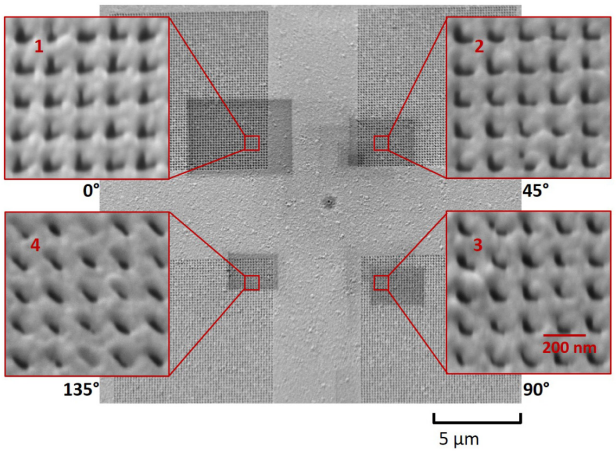

Fig. 2. V-shaped antenna based metasurface samples. The angles between the two arms of V-shaped antennas in the arrays of 1, 2, 3, 4 are 60°, 90°, 120°, 180° corresponding to the phase shift of 0°, 45°, 90°, 135°, respectively. A 30-nm-thick gold film was deposited on a glass substrate. Then, 30-nm-wide grooves were milled in this gold film to form the antennas. The arm lengths and the split angles between the arms were varied to introduce 45 degrees phase changes between different parts of the sample. The scale bar of 200nm is the same in all the insets.

refraction with a specific phase shift created by the metasurface has perpendicular polarization with respect to the incident beam. Therefore, a polarizer (not shown in Fig. 1) is used to filter the remaining part and keep the anomalously refracted beam (polarized along x) after the sample. A Spatial Light Modulator (SLM, Hamamatsu) is placed in the path of the detecting beam, transforming it into a vortex beam with orbital angular momentum of topological charge one. We also use a combination of a quarter wave plate and a polarizer to rotate the polarization of the detecting beam by 90 degrees and match it with the polarization of the abnormal refraction of the probing beam in order to enable formation of spiral interference in Fig. 3 on a CCD camera.

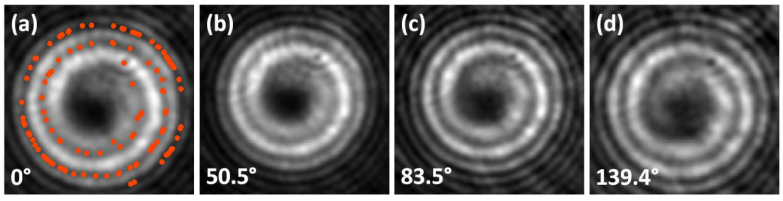

Fig. 3. Photos of the spiral interference patterns from the 4 samples: (a) sample 1; (b) sample 2; (c) sample 3; (d) sample 4. The dots in (a) indicate the location of numerically extracted minima. The size of each photo is 1.5mm × 1.5mm.

During the measurements, we focused the probing beam to the four samples subsequently. We obtained 40 interferograms registered for each sample and totally 160 photos for the 4 samples. Repositioning of the focused beam from one array to its neighboring array resulted in a rotation of the spiral interference pattern by approximately 45 degrees. Figure 3 shows one group of 4 photos for the 4 different antenna arrays. The rotation of the spiral patterns from Fig. 3(a) to (d) is in counterclockwise direction which agrees well with the design of the MSs, subsequently adding 45 degrees.

In order to obtain the phase shift from the experimentally measured interferograms, we need to calculate the rotation angles from the interference patterns obtained for the 4 samples. According to the method presented in there [40], the spiral interference pattern between a

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Gaussian beam and a vortex beam of charge one can be described by the following equation expressed in a polar coordinate (r, φ):

$$\varphi = ar^2 + b_0, \tag{1}$$

where a denotes the rate at which the spiral rotates as its radius increases, and is directly connected with the curvatures of the interfering wavefronts while b_0 is the starting phase of the spiral (spiral angle close to $r = 0$) showing the stating angle of spiral interference.

In all our measurements, the curvature of the interfering wavefronts was practically the same, regardless of the studied sample. The relative variation of the parameter a was smaller than 2%, which allowed us to assume that a was a constant. Therefore, the phase shift between each array can be accurately obtained by calculating the difference between the b_0 parameter retrieved from the interference patterns of the corresponding samples. In our analysis we digitalize the interferograms and extract the location of the intensity minima in the spiral pattern, as shown by the dots in Fig. 3(a). Then, we fit the location of the found minima using Eq. (1) with φ versus r^2, as shown in Fig. 4(a), and the parameter b_0 is extracted from the linear fit. We applied this fitting method to all 160 interferograms to calculate the corresponding values of b_0. The statistical analysis of the obtained data allowed us to retrieve the phase shifts introduced by the four MS samples that are graphically presented in Fig. 4(b). The measured phase shift agrees quite well with the phase shift intended by the MS design. The errors in the experimental measurements are mainly caused by the mechanical vibrations of the experimental setup.

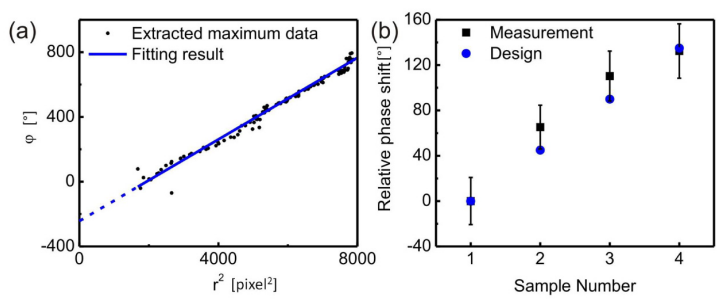

Fig. 4. Fitting result for the first sample (a) and phase shift results for all 4 samples (b).

4. Summary

In this paper, we report a reliable technique that is particularly useful for optical characterization of meta-structure phase properties using optical vortex interferometry. Phase shifts from four different metasurface samples are directly visualized by the proposed structured light based interferometric method. The measured results are in good agreement with the designed properties of the antennas arrays. There are several potential new capabilities that maybe enabled by the demonstrated technique. One of the advantages of the proposed structured light based interferometry is that it can be used to probe small fractions of the sample in a scanning fashion and as a result to map the phase shifts introduced by the regions that are much smaller that the incident beam diameter. This unique property is enabled by the presence of singularity in the detecting beam. Moreover, in addition to direct characterization of the metamaterials samples, the proposed technique can potentially be used for finding both the amplitude and phase information from the images.

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